



AM-BQA: Enhancing blind image quality assessment using attention retractable features and multi-dimensional learning

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ABSTRACT

In the realm of no-reference image quality assessment (NR-IQA), acquiring pristine source content for reference is often unattainable. This absence of reference presents challenges in accurately estimating perceptual scores due to the diversity and complexity of distortion patterns. To tackle these challenges, we introduce a novel approach named AM-BQA: Enhancing Blind Image Quality Assessment using Attention Retractable Features and Multi-Dimensional Learning, designed to capture and analyze complex patterns. Our method involves several steps. Firstly, we extract crucial and intricate features using a vision transformer. Next, we employ a multi-head transpose attention block with a dual key, incorporating overlap convolution patches and transpose attention into these extracted features. Finally, the attention maps generated by this process pass through an attention retractable block and a weighted multi-head layer to calculate the final quality score. By employing this architecture, we enhance both global and local interactions between complex patches. To validate the effectiveness of our approach, we assess it on four standard datasets (LIVE, TID2013, CSIQ, and KADID-10 K), including both synthetic datasets. Additionally, we conduct experiments on authentic datasets and demonstrate that our model achieves state-of-the-art performance across multiple datasets. The source code and pretrained models are available on this GitHub repository: <https://github.com/adhikariastha5/AM-BQA>.

1. Introduction

The ubiquity of smartphones has triggered a surge in daily digital image capture. However, these images often undergo operations such as compression, storage, transmission, and restoration during transfer between devices, leading to degraded quality. Accurate image quality assessment is vital in applications like security, surveillance, and medical imaging, directly impacting efficiency, dependability, and human lives. To address this, a valid and practical metric is needed to evaluate image quality across various distortions in real world scenarios. Objective metrics, including Full-Reference (FR-IQA), Reduced-Reference (RR-IQA), and No-Reference (NR-IQA) methods, utilize computer vision techniques to automatically learn the relationship between images and human-perceived quality. However, NR-IQA or Blind Image Quality Assessment [29] faces challenges due to the absence of reference images. Deep learning techniques have been employed to overcome this limitation, enabling flexible and adaptable approaches that extract relevant features to accurately capture complex distortions.

Effectively training deep learning models faces challenges due to

limited labeled data and the size of BIQA datasets with numerous trainable parameters. Researchers address these hurdles by incorporating subjective evaluations through MOS techniques. Humans focus on particulars, while machines emphasize distinctions between original and altered images as illustrated in Fig. 1. The potential for varied machine interpretations leading to misclassifications highlights the need to recognize limitations influenced by individual biases. This approach may lack validity in scenarios involving realistic distortions. Some methods incorporate knowledge from pretrained datasets [8] [40] or apply techniques such as meta-learning [42] and contrastive learning [19]. TReS [7] employs relative ranking to compare image quality, while MANIQA [33] deploys attention mechanisms to identify vital image regions without explicit labels.

The objective of this paper is to explore a novel approach to multi-dimensional learning, albeit with a nuanced perspective. We pose a fundamental question: Can the field of Blind Image Quality Assessment (BIQA) gain advantages from the additional knowledge acquired through multi-dimensional learning paradigms? Given that humans effortlessly discern images despite their distortions, as depicted in the

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varying viewpoints represented in the Fig. 1, it becomes imperative to develop methodologies that share a similar purpose – namely, to be resilient against diverse distortions and prioritize intricate patterns that remain insensitive to such distortions. This prompts the need to design techniques that can robustly evaluate image quality in the presence of varying distortions while simultaneously focusing on capturing intricate and distortion-insensitive features, akin to the way humans perceive images.

Built on recent progress in image restoration with attention retractable transformers [37], we introduce a novel multi-dimensional approach that focuses on learning intricate patterns while highlighting the content image. This technique effectively addresses the limitations associated with human scores and exploits supplementary insights from pretrained vision tasks. Our methodology incorporates the utilization of both pretrained weights and diverse attention mechanisms to unveil complex distortions, with a specific focus on local and global features. Primarily, our proposed AM-BQA architecture comprises four integral components. Firstly, an ImageNet pretrained Vision Transformer (ViT) [4] serves as the foundation for extracting both distorted types and pertinent features from input images. This allows the input image to be divided into patches, which are subsequently processed to extract diverse distortions and essential information. Secondly, an attention block, proposed by [36], enhances focus on relevant features, coupled with an attention retractable Transformer (ART) block amplifying local interactions. Furthermore, an adeptly designed weighted Multi-Layer Perceptron (MLP) header computes the ultimate quality score, efficiently assessing the significance of each patch. AM-BQA excels in precise noise distortion detection, meticulous local spatial interaction modeling, and dependable quality evaluations across an array of applications.

This paper introduces several notable enhancements:

- 1) Introducing a novel multi-dimensional deep learning methodology centered on capturing intricate patterns while accentuating image content. By tackling constraints linked to human assessments, our approach harnesses insights from pretrained vision tasks and utilizes varied attention mechanisms to reveal intricate distortions.
- 2) The utilization of dual key attention in combination with the Multi-Head Transpose Attention block. This integration effectively captures both local and global relationships within image features, enhancing the model's ability to understand complex patterns.
- 3) The application of the attention retractable transformer, which integrates contextual information and long-range dependencies. This

incorporation enables the model to capture and leverage crucial information for precise quality assessment.

- 4) The suggested framework demonstrates improvements across four different artificially altered datasets as well as some genuine distorted datasets, outperforming the capabilities of current cutting-edge models. This highlights the effectiveness and superiority of the developed method in improving blind assessment of image quality across a range of evaluation scenarios.

2. Related works

2.1. No-reference image quality assessment

Image quality assessment is crucial in scenarios where ground truth data is unavailable, necessitating the use of no-reference image quality assessment (NR-IQA) techniques. These methods, relying solely on the distorted image input, traditionally represent it with handcrafted features. Features rely on the assumption that natural scene statistics (NSS), exhibiting high regularity, aid in detecting artificial distortions such as spatial [21] [38], gradient [20], and wavelet based distortions [22]. However, real world distortions differ from laboratory simulated synthetic distortions, posing a challenge for traditional handcrafted features. Deep learning, particularly convolutional neural networks (CNNs), has revolutionized NR-IQA techniques, overcoming these limitations. CNN-based techniques employ CNNs to extract detailed image features [1] [18] [27] [31]. Early CNN-based NR-IQA approaches use pretrained CNN models [2] [9] to extract deep features from distorted images. These models, initially trained on large scale datasets for general computer vision tasks, learn complex representations applicable to image quality assessment. The NR-IQA field has witnessed the emergence of novel strategies. Hyper-IQA [25] introduced a feature separation method to distinguish between low level and high level features extracted from images. Meta-IQA [42] employed meta-learning techniques, training the network on various distortion types to provide the background knowledge necessary for distinguishing diverse distortions. Hallucinated-IQA [13] used a generative network to create reference images closely resembling actual images, focusing on estimating the perceptual quality of images through comparative analysis. Re-IQA leveraged unsupervised pre-training of high-level representations through the MoCo-v2 framework [24]. LIQE employed contrastive language-image pre-training (CLIP) to acquire both visual and textual embeddings [41]. The study by Zhang et al. facilitated model robustness to subpopulation shift [39], while Lifelong IQA employed a Split-and-

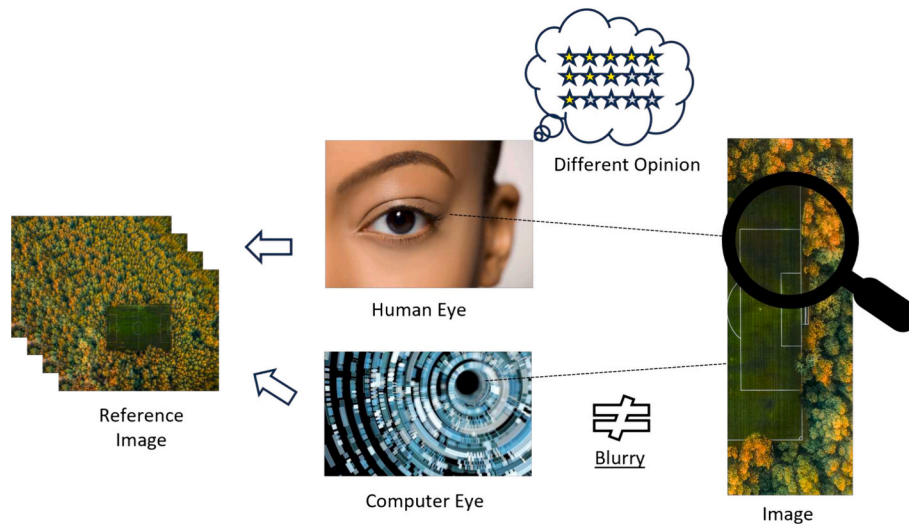


Fig. 1. The motivation of AM-BQA. Individuals possess varying viewpoints when it comes to images, whereas computers assess them in relation to a reference image. Furthermore, humans can perceive content even in the presence of blurriness.

Merge strategy to address catastrophic forgetting [15]. Additionally, Graph IQA investigated the relationship between distortion-related factors using graph representation [26].

2.2. Transformer in image quality assessment

CNNs, often used for FR-IQA and NR-IQA tasks, face challenges in capturing non-local features and integrating information across all image regions. This limitation hampers their ability to manage intricate feature combinations and detect broader interdependencies [7]. Additionally, challenges like spatial translation invariance and strong locality bias further hinder their performance. To address these limitations, researchers have turned to transformer models [28], initially designed for NLP tasks, as a solution. The Vision Transformer (ViT) [4] stands out for using multi-head attention to capture intricate connections in input data, surpassing the limitations of CNNs. In the field of IQA, IQT [35] combines a Transformer model for quality prediction with CNN-derived features extracted from both reference and distortion images. TransLA introduces an auxiliary network to predict salience maps, enhancing focus on crucial areas. MUSIQ [10] introduces a multi-scale IQA transformer to assess quality across various levels of granularity. RTN [17] used “progressive reinforced learning based instance discarding strategy” (PRID) to identify and discard less informative parts. TReS [7] employs relative ranking and self-consistency loss to enhance network robustness and exploit self-supervised information. Similarly, MANIQA [33] integrates attention mechanisms and pretrained transformers to identify pivotal image regions. In contrast to these approaches, our study considers content irrespective of distortion types, identifying intricate patterns through a multi-dimensional approach to transfer learning. This is achieved through a combination of specific, dense, and sparse attention mechanisms, facilitating the capture of both local and global receptive fields.

2.3. Attention mechanism

Attention mechanisms, widely embraced in various computer vision tasks, have garnered significant popularity [3] [14]. In the realm of No-Reference Image Quality Assessment (NR-IQA), researchers have explored inventive strategies to tackle computational challenges and incorporate global features. Restormer [36] introduces a groundbreaking strategy wherein attention mechanisms rearrange the spatial and channel dimensions within the network architecture. This ingenious approach seamlessly integrates global features into the model architecture while improving computational efficiency. In NR-IQA, Yang et al. [32] propose a saliency-guided architecture that employs both transposed and spatial attention mechanisms. The transposed attention mechanism considers contextual cues from various spatial locations, enabling the model to focus on specific image regions crucial for quality assessment. Like the self-attention blocks in the Swin Transformer [16], ART [37] uses both dense attention from local windows and sparse attention from sparse grids. This shift focuses on pixel-level information rather than semantic-level details. Yang et al. enhance the model’s capability to grasp essential visual intricacies and make accurate quality predictions by synergizing these attention mechanisms. These innovative concepts have inspired the current approach, which integrates the pixel-level insights from ART and a dual-key attention block. This amalgamation achieves weighted attention across various features in the feature map or input image, effectively capturing feature interdependencies.

3. Proposed method

Within this segment, we elaborate on our novel model, AM-BQA, which is a NR-IQA method incorporating essential components such as the Multi-head Dual Transpose Attention Block, Attention Retractable Transformer block, and Weighted Quality Score. This demonstrates the

benefits of employing multidimensional learning techniques. As depicted in Fig. 2, our method’s framework is presented. The primary objective of AM-BQA is to derive image features without the need for explicit labels, effectively discerning between essential content and intricate distortions.

3.1. Architecture overview

Given a provided distorted input image, denoted as $I \in \mathbb{R}^{H \times W \times C}$, where H , W , and C represent the image’s height, width, and channel count, our objective centers around evaluating a quality score ($\hat{q}$). While resizing images to smaller resolutions, such as 224×224 , may lead to degradation in quality due to loss of fine details and clarity, we address this challenge by incorporating a Vision Transformer (ViT) [4] trained via transfer learning for various visual tasks. The ViT serves as a feature extractor $f_{0/}$, where $0/$ represents a set of learnable parameters capable of discerning and classifying distortions within the provided image. Despite the reduction in resolution, the attention mechanism within the ViT preserves important spatial relationships and contextual information. Leveraging pre-trained transformer models, initially trained on large-scale image datasets like ImageNet, provides an advantage by encoding valuable knowledge in its parameters, thereby enhancing our ability to evaluate the quality of distorted images effectively.

The features extracted from the i th layer, with $i \in \{7, 8, 9, 10\}$ out of the 12 pretrained ViT layers, are represented by $F_i \in \mathbb{R}^{b \times c_i \times H_i \times W_i}$. These extracted features undergo further processing, including passing through the Attention Map ($A_{0/}$), Dense-Sparse block, and subsequently through the Weighted Quality Score module to compute the ultimate approximated quality score, $\hat{q}$. This approach leverages multidimensional learning by utilizing features from multiple layers to capture diverse aspects of image content, enhancing the model’s understanding of image characteristics for more robust and accurate quality assessment.

3.2. Multi-head dual transpose attention block

The self-attention layer within Transformers assumes a pivotal role in encompassing far-reaching relationships, pinpointing pertinent data, enabling parallel processing, modeling contextual dependencies, and flexibly handling diverse sequence dependencies [28]. In stark contrast to the traditional Transformer’s solitary key (K) computation, the proposed approach employs dual keys ($K = K1 + K2$) to augment the correlation between query (Q) and value (V). Additionally, rather than being confined solely to spatial dimensions, self-attention operates across channels as well. This entails computing cross-covariance spanning channels and utilizing transpose attention to generate an attention map that implicitly encodes the global context.

Moreover, by integrating depth-wise convolutions to emphasize local context, the technique heightens attention efficacy when generating the global attention map via feature covariance computation. In conjunction with the attention mechanism, the method introduces a modified variant of the feed-forward network (FN), referred to as overlap convolution. Unlike the conventional FN that individually processes each pixel location, the overlap patch mechanism employs two 1×1 convolutions to modify the features. Subsequently, the resultant features undergo a depth-wise convolution, capturing insights from adjacent pixel positions within a channel-wise spatial context. This facilitates the assimilation of local context during the learning process.

In Fig. 3, the multi-head dual transpose attention block operates on the combined feature $\tilde{F}_i \in \mathbb{R}^{b \times \sum c_i \times H_i \times W_i}$. This block takes this feature as its input. The block produces query (Q), key1 ($K1$), key2 ($K2$), and value (V) components using two independent overlap convolution transformations. These transformations are applied to a layer-normalized tensor $Y \in \mathbb{R}^{H \times W \times C}$. Specifically, a 1×1 convolution ($W_p^{(\cdot)}$)

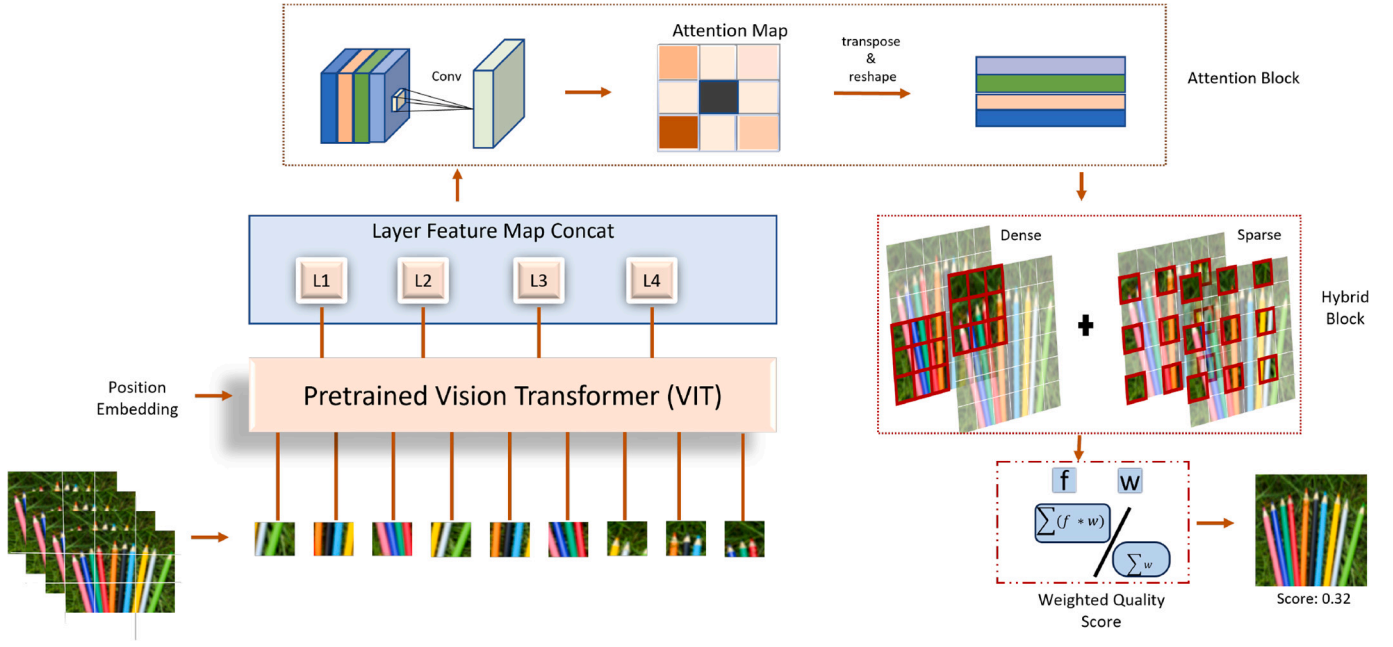


Fig. 2. AM-BQA: Architecture for Assessing Image Quality. The design consists of four elements: a pretrained Vision Transformer serving as a feature extractor, a Multi-scale Attention Block, a Hybrid Block, and a Weighted Quality Score.

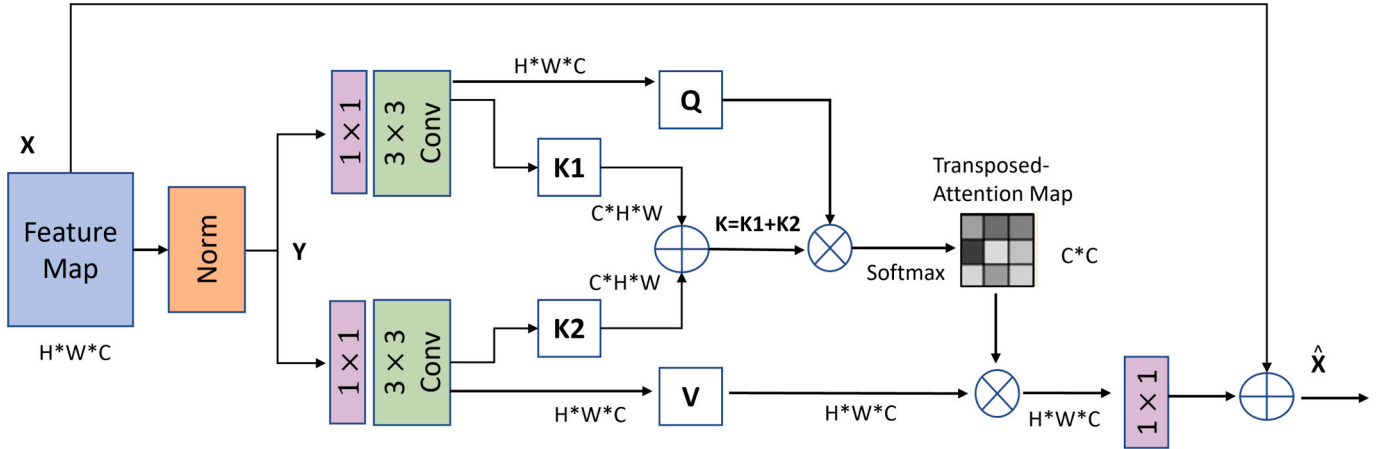


Fig. 3. A closer look at the Multi-Head Dual key Transpose Attention Block, where we use a dual key system to capture features dynamically augmented by a transposed mechanism along with convolution layers instead of basic forward layers, aiming to improve how features are extracted.

addresses pixel-level cross-channel context, while a 3×3 depth-wise convolution ($W_d^{(c)}$) handles channel-wise spatial context.

$$Q \& K1 = W_d^1 W_p^1 Y \quad (1)$$

$$K2 \& V = W_d^2 W_p^2 Y \quad (2)$$

Subsequently, the two keys are amalgamated, yielding $K = K1 + K2$. The query and key undergo reshaping to generate a transposed-attention map A of dimensions $\mathbb{R}^{C \times C}$, deviating from the conventional $\mathbb{R}^{H \times W \times H \times W}$. The ultimate output of attention from this block is represented as $\hat{X}$, calculated as follows:

$$\hat{X} = W_p \text{Attention}(Q, K = K1 + K2, V) + X \quad (3)$$

$$\text{Attention}(Q, K = K1 + K2, V) = (V) \cdot \text{Softmax}(K' \cdot (Q') / \alpha) \quad (4)$$

Here, both X and $\hat{X}$ represent the input (denoted as $\tilde{F}_i$) and the output feature maps, respectively. The matrices $\hat{Q} \in \mathbb{R}^{H \times W \times C}$; $\hat{K}1$ and

$\hat{K}2 \in \mathbb{R}^{C \times H \times W}$; and $\hat{V} \in \mathbb{R}^{H \times W \times C}$ are obtained through reshaping using the transposed attention technique. It's worth noting that C is numerically equivalent to c_i . The parameter α (representing spatial dimension) is a learnable scaling factor used to regulate the magnitude of the dot product prior to the application of the SoftMax function.

3.3. Attention retractable transformer block

The enhanced version of the Retractable Attention mechanism [37] implemented in this module facilitates the interaction of tokens, not only in densely populated regions but also in sparse areas of an image. This expansion of the receptive field is demonstrated in Fig. 4. Similar to the previous strategy of non-overlapping windows, this dense attention approach permits each token to engage with a smaller set of tokens from its neighboring non-overlapping $W \times W$ window. These token groups contribute to the computation of self-attention, referred to as D-MSA, as depicted in the figure. In contrast, the sparse attention mode enables each token to interact with a limited number of tokens located at sparse

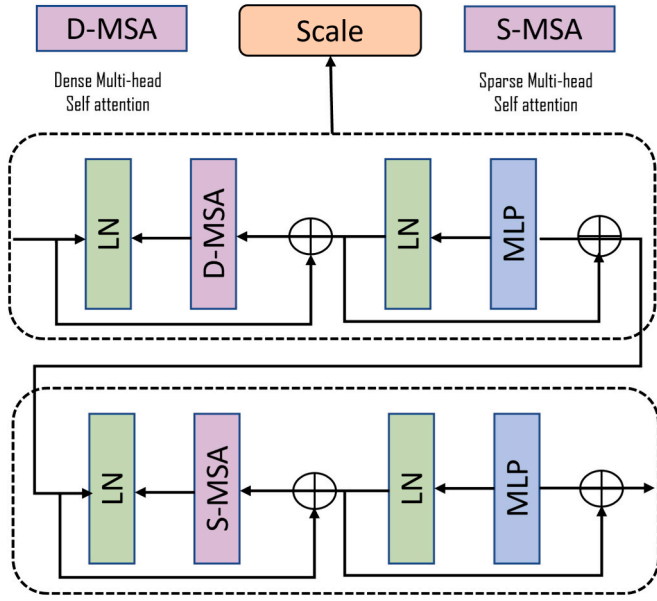


Fig. 4. In-depth explanation of the Attention Retractable Transformer architecture. In this design, we incorporate both dense and sparse multi-head self-attention mechanisms, rather than relying solely on a single type of multi-head attention.

positions with an interval I . These tokens are then divided into smaller clusters for self-attention computation, resulting in S-MSA. Both the D-MSA and S-MSA modules are computationally more efficient compared to the well-known Swin Transformer. The integration of these attention mechanisms enhances the model's ability to capture intricate low vision patterns and contextual dependency features.

3.4. Weighted quality score

The module calculates a comprehensive weighted patch score based on the processed feature acquired from preceding steps. Instead of relying on a solitary linear projection, two independent projections are employed: one for evaluating the patches (s_i) and another for assigning weights to the patches (w_i). The ultimate score is determined by multiplying these two projections, yielding a weighted average. This approach ensures an impartial evaluation of image quality, regardless of the involved regions. The final weighted sum average is computed as:

$$\hat{q} = \frac{\sum_{0 < i < N} w_i \cdot s_i}{\sum_{0 < i < N} w_i} \quad (5)$$

where N represents the number of patches within a single image.

4. Experiment

4.1. Experimental setup

4.1.1. Datasets

We evaluate the efficacy of the proposed framework using four synthetic distortion datasets and four authentic distortion datasets commonly utilized in real-world scenarios. These datasets provide a diverse range of distortions that mimic both synthetic and real-world conditions, ensuring a comprehensive evaluation of our framework's performance. LIVE [1], CSIQ [11], TID2013 [23], and KADID-10 K [12] comprise synthetic distortions, each offering unique challenges for image quality assessment algorithms. Notably, the KADID-10 K dataset stands out for its extensive variety of distortion types and corresponding distorted images. Among the authentic distortion datasets are CLIVE [6], KONIQ-10 K [8], SPAQ [5], which offers a rich collection of 11,125

images captured from smartphones, providing a real-world perspective on image quality assessment. A summary of the datasets utilized in our experiments is presented in Table 1.

4.1.2. Evaluation metrics

Similar to other advanced techniques, we assess the effectiveness of the proposed approach using SROCC (Spearman's rank-order correlation coefficient) and PLCC (Pearson's linear correlation coefficient). These metrics, which vary between -1 and 1 , act as evaluative standards for gauging the method's performance. Elevated values of both SROCC and PLCC, falling within the range of 0 to 1 , indicate heightened efficacy. Root Mean Square Error (RMSE) [30] serves as a prevalent method for quantifying disparities between predicted and actual values in perception score quality assessment. It assesses the magnitude of errors, offering a precise gauge of accuracy crucial for discerning discrepancies in forecasting errors across various estimators pertaining to a specific perceptual attribute.

The Pearson's Linear Correlation Coefficient (PLCC) is defined as follows:

$$PLCC = \frac{\sum_{i=1}^N (s_i - \mu_{s_i})(\hat{s}_i - \mu_{\hat{s}_i})}{\sqrt{\sum_{i=1}^N (s_i - \mu_{s_i})^2} \sqrt{\sum_{i=1}^N (\hat{s}_i - \mu_{\hat{s}_i})^2}} \quad (4.6)$$

Here, in the context of the i -th image, s_i represents the ground-truth quality score, while $\hat{s}_i$ represents the predicted quality score. The terms μ_{s_i} and $\mu_{\hat{s}_i}$ signify their respective means. The symbol N denotes the number of testing images.

The Spearman's Rank-Order Correlation Coefficient (SROCC) is defined as:

$$SROCC = 1 - \frac{6 \sum_{i=1}^N d_i^2}{N(N^2 - 1)} \quad (4.7)$$

Here, d_i signifies the difference between the ranks of the i -th test image in the ground truth and predicted quality scores.

Let us suppose that $\hat{\theta}$ be an estimator with respect to a given estimated parameter θ , the Root Mean Square Error is actually the square root of the Mean Square Error as

$$RMSE(\hat{\theta}) = \sqrt{MSE(\hat{\theta})} \quad (4.8)$$

4.1.3. Implementation details

In the proposed architecture, ViT acts as the feature extractor. Input patches are prepared for ViT through random cropping, yielding a consistent size of 224×224 . The ViT-B/8 [4] variant is chosen, pre-trained on ImageNet-21 k and fine-tuned on ImageNet1k [4], with an 8-patch size. The attention block, involving overlap patches, feeds into the Attention Retractable Transformer Block with an embedding dimension of 768 and a window size of 7. Following the standard strategy in IQA algorithms, the dataset is split randomly into training and testing sets, maintaining an 8:2 ratio. This is repeated five times with different seeds for distinct train-test splits. Approximately 80% of the data is for training, leaving 20% for testing. Experiments are conducted using

Table 1

Overview of well-known IQA datasets encompassing both artificially generated and real-world distortions, including their origins, the overall count of distorted images, and the variety of distortion types present in the synthetic datasets.

Dataset	Source	Images	Types	Type
LIVE	Kodak	779	5	Synthetic
CSIQ	Kodak	866	6	Synthetic
TID2013	Kodak	3000	24	Synthetic
KADID-10 K	Social Media	10,125	25	Synthetic
CLIVE	Mobile	1162	–	Authentic
KONIQ-10 K	Camera	10,073	–	Authentic
SPAQ	Smart-phones	11,125	–	Authentic

PyTorch 1.8.0 on a single RTX Titan GPU with CUDA 11.2. Training employs a batch size of 8 and the ADAM optimizer, using Mean Square Error (MSE) as the loss function. In testing, the experiment is repeated five times with varying seeds, and the final score is the average of metrics across outcomes.

4.2. Results

Table 2 offers a detailed analysis of the proposed method's performance on synthetic datasets. The AM-BQA consistently outperforms other models across all datasets in terms of image quality assessment (IQA). Across different synthetic datasets including LIVE, CSIQ, TID2013, and KADID-10 K, the AM-BQA model achieves the highest PLCC (Pearson Linear Correlation Coefficient) values and SROCC (Spearman Rank-Order Correlation Coefficient) values. Specifically, AM-BQA achieves outstanding PLCC and SROCC scores of 0.988 and 0.986 for LIVE, 0.974 and 0.968 for CSIQ, 0.948 and 0.941 for TID2013, and 0.951 and 0.946 for KADID-10 K, respectively. The second-best performing model varies depending on the dataset. However, when comparing across all datasets, there is a significant performance gap between AM-BQA and other state-of-the-art architectures. This emphasizes the robustness and effectiveness of AM-BQA in accurately assessing image quality across a variety of synthetic distortion datasets.

Table 3 illustrates the efficacy of our approach on authentic distortion datasets such as CLIVE, KONIQ-10 K, and SPAQ, highlighting its competence across well-established image quality benchmarks. AM-BQA, our proposed IQA model, demonstrates noteworthy superiority across all evaluated datasets. Specifically, on the CLIVE dataset, AM-BQA achieves a Pearson Linear Correlation Coefficient (PLCC) of 0.932 and a Spearman Rank-Order Correlation Coefficient (SROCC) of 0.926, outperforming established benchmarks such as DIVINE, CONTRIQUE, Re-IQA, LIQE, and TRes. This trend persists on the KONIQ-10 K dataset, where AM-BQA attains a PLCC of 0.936 and an SROCC of 0.932. Moreover, on the SPAQ dataset, AM-BQA exhibits unparalleled performance, registering a PLCC of 0.962 and an SROCC of 0.956. These results underscore the robustness and efficacy of the AM-BQA framework in discerning image quality nuances across varied datasets. Further, it underscores the versatility of our proposed approach in tackling various real-world image quality assessment challenges.

Within Table 4, a cross-dataset evaluation strategy is employed to assess the performance of various image quality assessment (IQA) methodologies, including our proposed AM-BQA, against competing models. Notably, training is conducted exclusively on the PIPAL dataset, ensuring a consistent framework, while testing unfolds across disparate datasets—LIVE and TID2013—without any adjustments or parameter adaptations. For synthetic image datasets, namely LIVE, CSIQ, and

Table 3

Comparative Evaluation of Our Proposed Method's Performance on Authentic Distortion Datasets alongside State-of-the-Art Models.

Method	CLIVE		KONIQ-10 K		SPAQ	
	PLCC	SROCC	PLCC	SROCC	PLCC	SROCC
DIVINE [22]	0.591	0.588	0.558	0.546	0.600	0.599
CONTRIQUE [19]	0.845	0.857	0.906	0.894	0.919	0.914
Re-IQA [24]	0.854	0.840	0.923	0.914	0.925	0.918
LIQE [41]	0.910	0.904	0.908	0.919	–	–
TRes [7]	0.877	0.846	0.928	0.915	–	–
AM-BQA (Ours)	0.932	0.926	0.936	0.932	0.962	0.956

Table 4

Cross dataset evaluation. PIPAL is used as the training dataset. Lower RMSE shows better performance, while higher PLCC and SROCC scores mean better results.

Train On	PIPAL					
	LIVE			TID2013		
	PLCC	RMSE	SROCC	PLCC	RMSE	SROCC
P2P-BM [34]	0.386	–	0.424	0.428	–	0.426
HyperIQA [25]	0.426	–	0.486	0.414	–	0.384
TRes [7]	0.643	0.466	0.663	0.516	0.482	0.563
MANIQA [33]	0.835	0.342	0.855	0.704	0.396	0.619
AM-BQA (Ours)	0.841	0.326	0.864	0.726	0.389	0.694

TID2013, we focus on four shared distortion types: JPEG, JPEG2K, WN, and Blur. From the table, it is evident that AM-BQA exhibits remarkable performance across both LIVE and TID2013 datasets when compared to alternative IQA methods. Specifically, on the LIVE dataset, AM-BQA achieves a PLCC of 0.841 and a SROCC of 0.864. These scores surpass those of competing methodologies such as P2P-BM, HyperIQA, TRes, and MANIQA, indicating AM-BQA's superior ability to accurately predict image quality. Similarly, on the TID2013 dataset, AM-BQA continues to demonstrate its robustness, recording a PLCC of 0.726 and an SROCC of 0.694. Again, these scores outshine those of the comparative methods, underlining the consistent performance superiority of AM-BQA across diverse datasets. Additionally, AM-BQA achieves a notable RMSE of 0.326 on the LIVE dataset and 0.389 on the TID2013 dataset, further corroborating its efficacy in accurately estimating image quality.

In Table 5, a comprehensive comparison is presented showcasing the performance of our method alongside other cutting-edge models on the LIVE dataset, focusing on various distortion types. This thorough examination sheds light on how our approach fares against existing techniques in handling diverse image distortions. Notably, our method, AM-

Table 2

Comparative Analysis of Our Proposed Method Against Previous State-of-the-Art Architectures on Synthetic Distortion Datasets.

Dataset	LIVE		CSIQ		TID2013		KADID-10 K	
Criterion	PLCC	SROCC	PLCC	SROCC	PLCC	SROCC	PLCC	SROCC
DIVINE [22]	0.908	0.892	0.776	0.804	0.567	0.643	0.435	0.413
MEON [18]	0.955	0.951	0.864	0.852	0.824	0.808	0.691	0.604
WaDIQaM [2]	0.955	0.960	0.844	0.852	0.855	0.835	0.752	0.739
DBCNN [40]	0.971	0.968	0.959	0.946	0.865	0.816	0.856	0.851
TIQA [35]	0.965	0.949	0.838	0.825	0.858	0.846	0.855	0.850
MetaIQA [42]	0.959	0.960	0.908	0.899	0.868	0.856	0.775	0.762
CONTRIQUE [19]	0.961	0.960	0.955	0.942	0.857	0.843	0.937	0.934
Re-IQA [24]	0.971	0.970	0.960	0.947	0.861	0.804	0.885	0.872
LIQE [41]	0.951	0.970	0.939	0.936	–	–	0.931	0.930
Continual [39]	–	0.9602	–	0.8480	–	–	–	0.8801
Graph IQA [26]	0.980	0.979	0.959	0.947	–	–	–	–
P2P-BM [34]	0.958	0.959	0.902	0.899	0.856	0.862	0.849	0.840
HyperIQA [25]	0.966	0.962	0.942	0.923	0.858	0.840	0.845	0.852
TRes [7]	0.968	0.969	0.942	0.922	0.883	0.863	0.858	0.915
MANIQA [33]	0.983	0.982	0.968	0.961	0.943	0.937	0.946	0.944
AM-BQA (Ours)	0.988	0.986	0.974	0.968	0.948	0.941	0.951	0.946

Table 5
SROCC comparison on each distortion in LIVE dataset.

Method	JP2K	JPEG	WN	GB	FF
WaDIQaM [2]	0.942	0.953	0.982	0.938	0.923
HyperIQA [25]	0.949	0.961	0.982	0.926	0.936
DBCNN [40]	0.955	0.972	0.980	0.935	0.930
GraphIQA [26]	0.979	0.978	0.978	0.978	0.979
AM-BQA (Ours)	0.986	0.984	0.987	0.985	0.986

BQA, consistently surpasses other state-of-the-art models across all distortion categories. Specifically, it achieves the highest SROCC scores of 0.986 for JPEG2000 (JP2K), 0.984 for JPEG, 0.987 for white noise (WN), 0.985 for Gaussian blur (GB), and 0.986 for JPEG2000 with bit errors in simulated Rayleigh fading channel (FF). This robust performance underscores AM-BQA's efficacy in accurately assessing image quality across a spectrum of distortion types. Such reliability positions it as a superior choice for real-world applications in image quality assessment. The diversified set of distortions in the LIVE dataset ensures a comprehensive evaluation of image quality assessment methods across various realistic scenarios, thereby enhancing the credibility of the comparative analysis presented.

Additionally, Fig. 5 complements this analysis by providing a visual representation of distorted images alongside their corresponding attention maps, Mean Opinion Scores (MOS), and our predicted scores. This visualization not only facilitates a deeper understanding of the impact of different distortions on the attention map but also aids in interpreting the influence of these distortions on overall image quality assessment. Through these combined analyses, our method's robustness and effectiveness in accurately evaluating image quality across diverse distortion types are underscored, offering valuable insights. Furthermore, the figure highlights the correlation between the attention maps

and the MOS values, showcasing how the model's attention focuses on the regions of the image that are most affected by the distortions. The attention maps in the figure show us where the model is paying attention to in the distorted images when it's trying to guess how good or bad the pictures look. By looking at these attention maps, we can see which parts of the image the model thinks are most important for judging the quality. This helps us understand how the model works and what it's good at, or not so good at, when it comes to evaluating images with different kinds of distortions. The scatterplot in Fig. 6 demonstrates that the proposed AM-BQA model yields better results compared to previous state-of-the-art methods on the PIPAL datasets.

4.3. Ablation study

In this section, we delve into a thorough evaluation of the model's performance by implementing multiple alterations, aimed at elucidating the influence of various elements within the model. Leveraging the CSIQ and KADID-10 K datasets, we conduct experiments under consistent conditions, employing a fixed batch size of 8 and an initial learning rate of 0.003 across all trials. Tables 4 and 6 provide a detailed overview of the effects of individual blocks within our proposed method.

Our analysis emphasizes the multidimensionality of the evaluation process, aligning with the concept of multi-dimensional learning. By systematically scrutinizing four distinct constituents, such as the feature extractor and attention block, we aim to discern the significance of each block and rationalize requisite modifications. This approach allows us to gain valuable insights into the impact of components on scores and the overall model performance.

Explicitly highlighting the different dimensions or aspects of the model under examination enhances our understanding of its capabilities and limitations. By assessing how each component contributes to the

Distorted Images

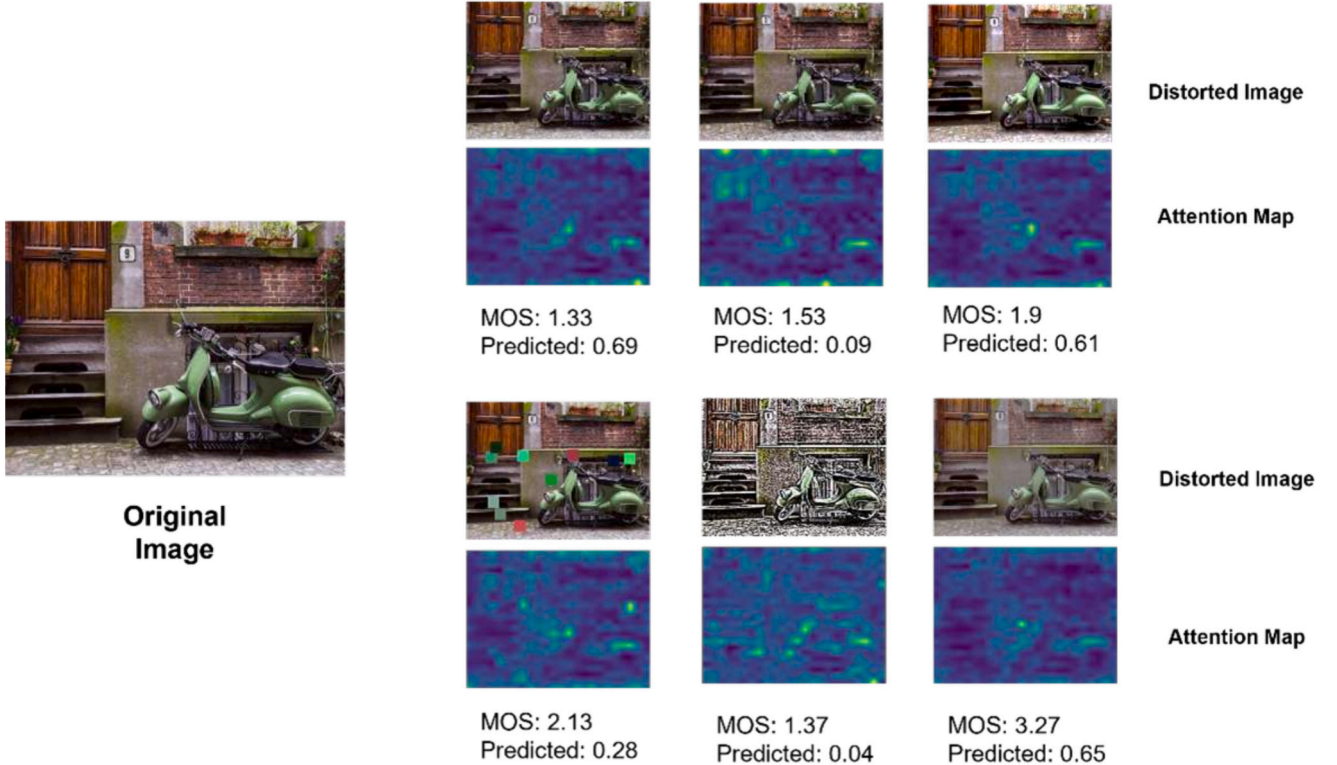


Fig. 5. Example images from the dataset. On the left side, we present the original image, while the right side showcases the distorted images along with the attention map, followed by the MOS scores and our predicted results. MOS scores represent the ground-truth human ratings.

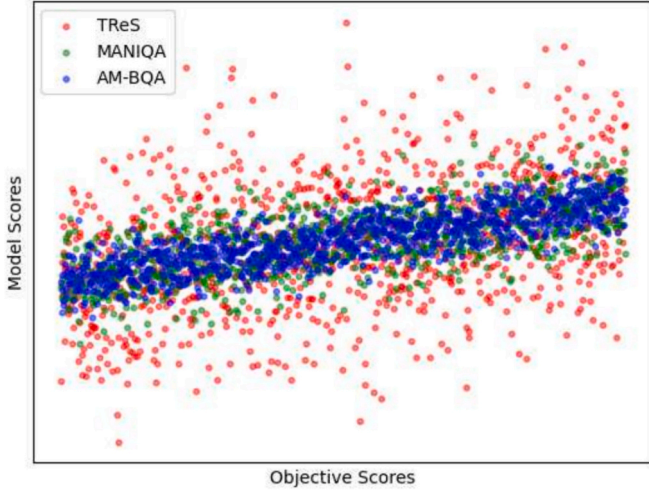


Fig. 6. Scatter plots comparing ground-truth mean opinion scores (MOSs) with predicted scores from the proposed AM-BQA model and other state-of-the-art models on PIPAL datasets.

Table 6

Examining the Impact of Various Blocks, including Attention, ART, and Weighted Blocks, on PIPAL Datasets: An Exploration of Multidimensional Learning.

	Attention Block	ART Block	Weighted Block	PLCC	SROCC
1				0.618	0.596
2	✓			0.648	0.614
3		✓		0.651	0.632
4	✓		✓	0.646	0.628
5		✓	✓	0.664	0.646
6	✓	✓	✓	0.724	0.692

model's efficacy, we identify avenues for enhancement and optimization. This analytical approach furnishes a comprehensive comprehension of the model's behavior and functionality, contributing to a more nuanced and insightful evaluation.

Feature Extractor. Table 6 illustrates a significant advantage in using ViT as the feature extractor compared to CNN-based networks, particularly Resnet50 in other state-of-the-art models. ViT achieves higher scores, emphasizing the importance of integrating transformers to effectively capture extensive dependencies in image analysis. In the domain of NR-IQA, pretrained models trained on smaller patches demonstrate heightened competitiveness compared to alternative approaches. These models, refined with smaller image patches, excel in assessing image quality within real-world scenarios, outperforming other methodologies in NR-IQA tasks.

Attention Block. The proposed attention block significantly enhances global interactions across all channels, facilitating comprehensive information exchange. The results in Table 6 not only validate the necessity of these interactions but also underscore their critical role in method efficacy. Comparing Table 4 and Table 6 highlights the substantial performance boost attained by incorporating transformer self-attention. Combining this attention module with ViT in Table 6 produces a greater score increase than using transformer attention alone with other CNN networks like Resnet50 in previous state-of-the-art models. Integrating this module with the other two modules notably improves both SROCC and PLCC metrics by 0.05, highlighting the attention block's crucial role in strengthening global interactions for better performance.

Furthermore, the detailed examination in Table 7 reveals the benefits of multi-head convolution compared to a standard linear layer, particularly in single and double key scenarios. The table provides compelling

Table 7

Assessing the Influence of Single or Dual Key Mechanisms within Linear or Multi-head Convolution on PIPAL Datasets: Unveiling the Effects of Multidimensional Learning.

	Single Key	Dual Key	PLCC	SROCC
Linear	✓		0.664	0.642
		✓	0.682	0.668
Convolution	✓		0.712	0.682
		✓	0.724	0.692

evidence of multi-head convolution's benefits, demonstrating superior performance and capabilities compared to linear layers. This underscores the importance of harnessing multi-head convolution's potential to maximize outcomes across diverse scenarios. By leveraging multi-head convolution, the model captures intricate patterns, extracts complex features, and enhances its predictive capacity. This highlights multi-head convolution's critical role in boosting the model's effectiveness, extending its performance boundaries, and ensuring superior results across applications.

Attention Retractable Transformer Block. The revised retractable attention block is designed to amplify local interaction through dense layers while also capturing complex interactions to attain global information using sparse layers. Incorporating this block yields noticeable improvements in both SROCC and PLCC metrics, as seen in Table 6. Furthermore, comparing the scores from this block to those from SWIN, as shown in Table 8, emphasizes improvements in both performance and computational efficiency. These findings underscore the significance of leveraging features from both sparse and dense perspectives, rather than solely focusing on dense interactions.

Weighted Quality Score. The Weighted Quality Score module introduced in this study is tailored to calculate the weighted score for each individual patch. Including this module significantly enhances performance, highlighting its effectiveness in capturing pertinent details through improved SROCC and PLCC values. Drawing inspiration from a prior model that employed triplet loss to encompass all features, this module was devised to incorporate comprehensive information. As shown in Table 6, the Weighted Quality Score module prominently elevates the model's performance. By effectively mitigating the impact of high-weight patches, the approach adeptly reduces the risk of overfitting. The Weighted Quality Score module plays a pivotal role in enhancing the model's overall robustness and its ability to generalize effectively.

4.4. Efficiency analysis

In contrast to the preceding state-of-the-art models, MANIQA and TReS, our model, AM-BQA, despite its robust architecture, demonstrates remarkable superiority. It achieves faster training convergence, requiring fewer epochs and less time to attain a performance comparable to its counterparts as seen in Table 9. Furthermore, AM-BQA exhibits an efficient use of computational resources, leading to reduced memory consumption during the training process.

5. Conclusion

We present AM-BQA, a new method that uses a multidimensional learning approach to analyze and capture complex patterns in images.

Table 8

Examining the Influence of ART Versus Swin Blocks: A Comparative Analysis of Their Effects on the score.

	PLCC	SROCC
ART [37]	0.724	0.692
Swin [16]	0.698	0.662

Table 9

Evaluating the Efficiency of Our Model Compared to State-of-the-Art Models in Terms of Parameter Count and Computational Complexity.

Model	Param	Flops
TReS [37]	350 M	71.82B
MANIQA [7]	240 M	48.42B
AM-BQA(Ours)	162 M	31.42B

This method considers multiple factors and aspects of an image when evaluating its quality, going beyond just looking at individual pixels or small parts of the image. It consists of four key components that address issues related to understanding the context of the image, dealing with long-range dependencies, considering spatial interactions, and handling channel feature interactions. The use of a multi-head transpose attention block, along with a dual-key mechanism, allows for the efficient identification and handling of irrelevant features within the image.

By taking into account various dimensions, such as color, texture, sharpness, and blurriness, AM-BQA provides a more comprehensive and accurate assessment of image quality. This is demonstrated through experiments conducted on four commonly used datasets that contain synthetically distorted images. The results show that our approach outperforms existing methods, showcasing its effectiveness in accurately evaluating image quality.

Looking to the future, researchers could explore ways to include descriptions of how subjects make their quality judgments in existing image quality assessment (IQA) datasets. This would help improve models' understanding of the human visual system (HVS) and the reasons behind quality scores, leading to more accurate assessments. It's worth noting that subjects may show less tolerance towards certain types of distortions, such as blur or grayscale images, but our model is able to distinguish between different perceptual cues and explain the variations in assigned scores. By employing a multidimensional learning approach, AM-BQA offers a more thorough and precise method for assessing image quality.

CRedit authorship contribution statement

Astha Adhikari: Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Methodology, Investigation, Conceptualization. **Sang-Woong Lee:** Writing – review & editing, Validation, Supervision, Project administration, Investigation, Funding acquisition.

Declaration of competing interest

None.

Data availability

No data was used for the research described in the article.

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